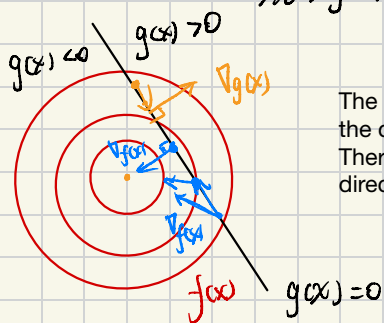


$$\{\pi_j^*\} = \arg \max_{\{\pi_j\}} \sum_{j=1}^K N_j \log \pi_j, \quad \text{s.t.} \quad \underbrace{\sum_{j=1}^K \pi_j = 1}_{\text{equality constraint}}, \quad \pi_j \geq 0$$

general: $\{x^*\} = \arg \max_x f(x) \leftarrow \text{obj. func}$
 s.t. $g(x) = 0$



The gradient of $g(x)$ is perpendicular to the constraint surface
 There is not gradient component in the direction of constraint surface.

The gradient of r is orthogonal to the constraint surface when $f(x)$ cannot increase without leaving the surface

$$\therefore \nabla f(x) \nparallel \nabla g(x) \Rightarrow \nabla f(x) + \lambda \cdot \nabla g(x) = 0$$

$$\exists \lambda \in \mathbb{R}, \lambda \neq 0$$

if $\lambda < 0$, $\nabla f(x)$ and $\nabla g(x)$ are in the same direction
 $\lambda > 0$ opposite

Define Lagrange

$$L(x, \lambda) = f(x) + \lambda \cdot g(x)$$

stationing point: $\frac{\partial}{\partial x} L(x, \lambda) = 0 \Rightarrow \text{optimal } x^*$
 $\frac{\partial}{\partial \lambda} L(x, \lambda) = 0 \Rightarrow \lambda^*$

$$f(\pi) = \sum_{j=1}^K N_j \log \pi_j \quad g(\pi) = \sum_{j=1}^K \pi_j - 1$$

$$L(\pi, \lambda) = \sum_{j=1}^K N_j \log \pi_j + \lambda \left(\sum_{j=1}^K \pi_j - 1 \right)$$

$$\frac{\partial}{\partial \pi_j} L(\pi, \lambda) = \frac{N_j}{\pi_j} + \lambda = 0 \Rightarrow N_j + \lambda \pi_j = 0 = \pi_j = \frac{-N_j}{\lambda} = \frac{N_j}{\sum_k N_k}$$

$$\frac{\partial}{\partial \lambda} L(\pi, \lambda) = \sum_{j=1}^K \pi_j - 1 = 0$$

$$\Rightarrow \sum_{j=1}^K N_j + \lambda \sum_{j=1}^K \pi_j = 0 \Rightarrow \lambda = -\frac{\sum_{j=1}^K N_j}{\sum_{j=1}^K \pi_j}$$

Tutorial 4 PS 4.6 Expectation Maximization for Mixture of Exponentials

$$p(x) = \sum_{j=1}^K \pi_j p(x|j), \quad p(x|j) = \lambda_j e^{-\lambda_j x}, \quad \lambda_j > 0$$

Given samples: $D = \{x_1, x_2, \dots, x_n\}$

Find $\theta = \{\pi_j, \lambda_j\}_{j=1}^K$

Joining LL

$$p(x_i, z_i) = \prod_{j=1}^K \pi_j^{z_{ij}} p(x_i|j)^{z_{ij}} \quad z_{ij} = \begin{cases} 1 \\ 0 \end{cases}, \pi_j \text{ (prob)}$$

$$\log p(x_i, z_i) = \sum_j z_{ij} \log \pi_j + z_{ij} \log p(x_i|j)$$

$$\log p(x, z) = \sum_i \sum_j z_{ij} \log \pi_j + z_{ij} \log p(x_i|j)$$

Z-step: $Q(\theta, \hat{\theta}^{old}) = E_{z|x, \hat{\theta}^{old}} [\log p(x, z)]$

$$= \sum_i \sum_j \pi_j^{z_{ij}} \log \pi_j + \pi_j^{z_{ij}} \log p(x_i|j)$$

$$\hat{z}_{ij} = E_{z|x} [z_{ij}] = p(z_{ij}=1|x) \cdot 1 + p(z_{ij}=0|x) \cdot 0 \quad \text{ps 4-6, Expectation of indicator variable}$$

$$= p(z_{ij}=1|x)$$

$$= \frac{p(x|z_{ij}=1) \cdot p(z_{ij}=1)}{p(x)}$$

recall: x_i 's independent: $p(x) = p(x_1) \cdot p(x_2) \cdot \dots \cdot p(x_n)$

$$= p(x_i) p(x_{-i})$$

all other x_i , not i

$$p(x_i | z_{ij}=1) = p(x_i | z_{ij}=1) \cdot p(x_{-i} | z_{ij}=1) \\ = p(x_i | z_{ij}=1) \cdot p(x_{-i})$$

$$\Rightarrow \hat{z}_{ij} = \frac{p(x_i|j) \cdot p(x_{-i}) \cdot p(z_{ij}=1)}{p(x_i) - p(x_i)} \\ = \frac{p(x_i|j) \cdot \pi_j}{\sum_k \pi_k p(x_i|k)} = p(z_{ij}=1|x)$$

M-step: $Q = \sum_i \hat{z}_{ij} \log \pi_j + \hat{z}_{ij} \log p(x_i|j)$

solve λ_j :

$$\frac{\partial}{\partial \lambda_j} Q = \sum_i \hat{z}_{ij} \frac{\partial}{\partial \lambda_j} p(x_i|j) = 0 \\ = \sum_i \hat{z}_{ij} \frac{\partial}{\partial \lambda_j} (\log \lambda_j - \lambda_j x_i) = 0 \\ = \sum_i \hat{z}_{ij} (\frac{1}{\lambda_j} - x_i) = 0$$

$$\lambda_j = \left(\frac{\sum_i \hat{z}_{ij} x_i}{\sum_i \hat{z}_{ij}} \right)^{-1}$$

Solve π_j

$$\{\pi_j^*\} = \arg \max_{\pi_j} Q = \arg \max_{\pi_j} \sum_i \hat{z}_{ij} \log \pi_j = \sum_j \log \pi_j \cdot \underbrace{\left(\sum_i \hat{z}_{ij} \right)}_{N_j \text{ (PS 4-12)}}$$

$$\Rightarrow \sum_j N_j \log \pi_j$$

PS 4-12 $\pi_j^* = \frac{N_j}{\sum_k N_k}$